

Regime-Dependent Ranking and Low-Cost I/Q Repair for Automatic Modulation Classification Under Structured Interference

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Abstract—Compact signal representations reduce the inference cost of learned wireless receivers, but their relative value can change with interference, mobility, and channel regime. This paper studies that instability for automatic modulation classification (AMC) under structured interference. A source-disjoint simulation campaign first shows that marginal scores conceal a severe-condition reversal between a compact spectral model and stronger I/Q-domain references. An independently generated severe-held evaluation then shows that the favorable campaign ordering and an overlap-SIR diagnostic surface do not transport uniformly to new jammer, mobility, and channel combinations. Spectral-width, clean-coverage, probe, and negative-control tests are consistent with a representation-level explanation for the exposed degradation. To preserve complementary information at low cost, we add a depthwise-separable received-I/Q sidecar to the compact spectral path. On the same severe-held benchmark, the sidecar improves macro-F1 by 4.37 percentage points with about 3% additional MACs. A retrained variant with permanently zeroed I/Q input returns to the spectral baseline, supporting the interpretation that informative received-I/Q samples, rather than added capacity or fusion structure alone, enable the repair. Absolute macro-F1 remains low in this extreme regime; the reported gain is therefore a robustness recovery, not evidence of deployment-level performance.

Index Terms—Automatic modulation classification, structured interference, rank instability, representation repair, domain shift, compact neural networks, interference overlap, vehicular Doppler.

I. INTRODUCTION

AUTOMATIC modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform’s modulation from received samples. Deep AMC has progressed from convolutional classifiers of in-phase and quadrature (I/Q) samples [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9]. Their efficiency, however, does not establish that a compressed representation remains preferable across the operating conditions encountered by a mobile receiver.

This limitation is consequential in mobile and vehicular receivers. Structured interference and mobility can change which signal representation is most useful under a fixed compute budget. A campaign-level average can then hide both a conditional rank reversal and the risk that the apparent favorable region will disappear under a new operating regime. Learned receivers therefore need more than a static leaderboard: they need evidence that representation rankings remain stable across relevant conditions, or a low-cost mechanism that retains the information required when a compact path fails.

This paper follows that requirement as a single evidence chain. We first test whether compact spectral and high-

capacity I/Q models retain the same ordering across structured-interference conditions. We then evaluate the campaign’s severe-condition ordering on independently generated jammer, mobility, and channel combinations. After the ordering fails to transport, targeted controls test whether the exposed degradation follows from spectral-model capacity, condition coverage, classification-head limitations, or missing input information. The final intervention adds a small received-I/Q path to the compact spectral receiver and tests whether complementary samples recover part of the degradation.

The compact starting point is VIMD-Net, which maps a complex short-time Fourier transform (STFT) to three allocation routes and uses a training-only simulation-component teacher. Its complete A5 configuration improves over the shared A0 spectral backbone by 4.57 percentage points, although the isolated teacher-form and route-count contrasts remain unresolved. This result establishes the compact spectral baseline; the paper’s central contribution is the ranking, transport, diagnosis, and repair evidence built around it.

The main contributions are:

- We establish condition-dependent model ranking over the complete SNR–SIR grid and show, on a source-disjoint independent severe-held set, that the campaign’s favorable compact-model ordering does not persist across new jammer, mobility, and channel combinations.
- We distinguish representation loss from alternative explanations with spectral-width, clean-coverage, frozen-probe, and route-gating controls. The combined evidence supports a representation-level limitation while retaining the stated threshold and joint-shift limitations.
- We introduce a lightweight received-I/Q sidecar that repairs 4.37 percentage points of macro-F1 on the severe-held benchmark at about 3% additional MACs. Matched capacity controls and a retrained zeroed-I/Q ablation support attribution to informative I/Q input rather than to added capacity or fusion structure alone.

II. RELATED WORK AND STUDY POSITIONING

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated reparameterized causal convolutions, IQ-imbalance- and carrier-offset-aware kernel selection, and channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate the strong I/Q-domain

comparators used here and position lightweight fusion as a repair mechanism rather than as a stand-alone architectural novelty.

B. Robust, Cross-Domain, and Interference-Aware AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16], while overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation populate the recent frontier [17]–[19]. Interference-tolerant [16] and tensor-based malicious-interference [20] recognition address structured interference directly. Our setting instead studies how the relative performance of compact spectral and high-capacity I/Q representations changes with interference condition and regime.

C. Gap and Study Positioning

Knowledge-guided and multimodal preprints continue these directions [21]–[24]. Ranking changes under distribution shift are a known broader phenomenon in robustness and benchmark studies [25]–[27]. Most AMC studies compare global accuracy, report degradation under a selected shift, or reduce the cost of one architecture. This work instead asks whether the relative ordering of spectral and I/Q representations is stable, whether an observed conditional advantage transports to a new regime, and whether a small complementary signal path can repair the resulting failure. A low-dimensional overlap–SIR surface is used only as a campaign diagnostic; its transfer is tested rather than assumed.

III. SIGNAL MODEL AND COMPACT SPECTRAL RECEIVER

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa \mathcal{H}_u(u)[n] + w[n]), \quad (1)$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [28] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10 \log_{10}(E[|\mathcal{H}_s(s)|^2]/E[|\kappa \mathcal{H}_u(u)|^2])$ on the simulator’s tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [29], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. The held-speed split probes deeper intra-window non-stationarity: $T_c \approx 0.423/f_D$ falls to about 0.31 ms at 250 km/h, below the 1024 μ s decision window.

Training pairs two independently impaired observations of the same immutable source; inference uses one view, so pairing is a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, swapping two admissible emitters can preserve the mixture while changing the requested label.

B. Three-Route Spectral Allocation

Let $Z = \text{STFT}(x) \in \mathbb{C}^{F \times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} [M_s, M_j, M_o] &= \text{softmax}([A_s, A_j, A_o]), \\ M_s + M_j + M_o &= 1. \end{aligned} \quad (2)$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ . The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \quad (3)$$

$$Z_j = (M_j + (1 - \lambda)M_o) \odot Z. \quad (4)$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range $[0.05, 0.35]$ and is added only to the modulation route; the retained weights therefore remain nonnegative and sum to $1 + \rho$. Equivalently, $Z_m + Z_j = (1 + \rho)Z$, rather than reconstructing the original mixture exactly. This is the ratio-masking construction familiar from supervised source separation [30]–[33]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student’s lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^* = [q_s - q_j]_+, \quad M_j^* = [q_j - q_s]_+, \quad (5)$$

$$M_o^* = q_u + 2 \min(q_s, q_j). \quad (6)$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

D. Training Objective

The total loss is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{mod}} + \alpha \mathcal{L}_{\text{teacher}} + \beta \mathcal{L}_{\text{jammer}} + \gamma \mathcal{L}_{\text{link}} \\ &\quad + \eta \mathcal{L}_{\text{contrast}} + \theta \mathcal{L}_{\text{orth}}, \end{aligned} \quad (7)$$

where $\mathcal{L}_{\text{mod}}$ is the modulation classification loss, $\mathcal{L}_{\text{teacher}}$ is a Jensen–Shannon [34] alignment between student and teacher

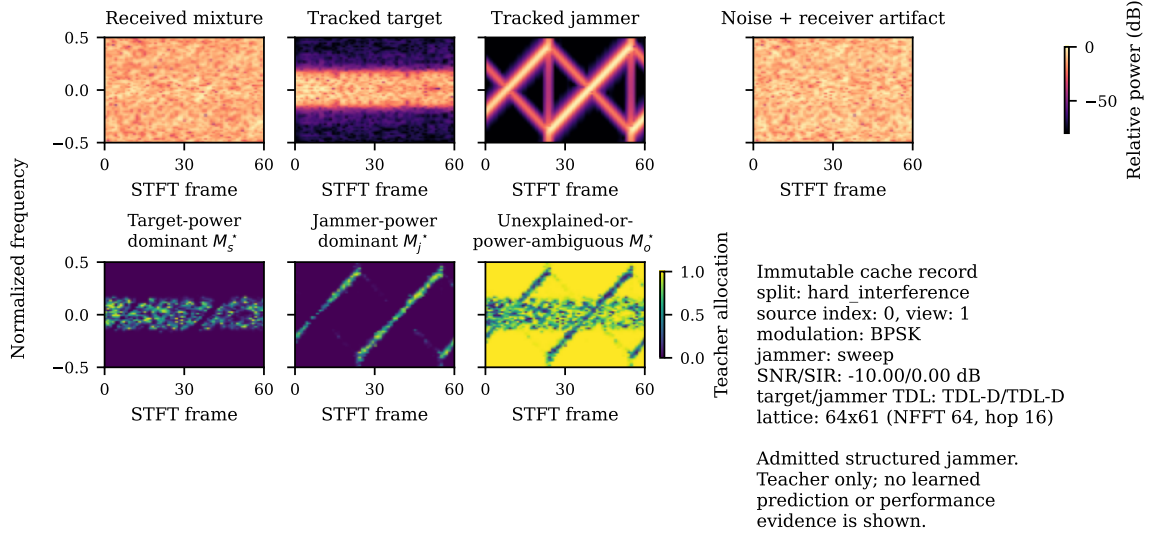


Fig. 1. Training-only supervision for the compact spectral routes. The top row shows the received mixture and simulator-tracked target, jammer, and residual components; the bottom row shows the corresponding target-dominant, jammer-dominant, and uncertainty masks from Eqs. (5)–(6). Only the received mixture is available at inference.

masks, $\mathcal{L}_{\text{jammer}}$ and $\mathcal{L}_{\text{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\text{contrast}}$ is a supervised-contrastive loss [35] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\text{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective.

E. Information Contract and Complexity

At inference every model receives only the mixture; simulator-tracked target, jammer, and SIR quantities are training-only or post-hoc audit variables. VIMD-Net uses 39500 parameters and 41.8 million reported MACs.

F. Implementation Details

Unless otherwise stated, all models share the same source-disjoint cache, data budget, optimization budget, stopping rule, and validation-only selection criterion; model-specific components follow each variant’s predefined configuration.

Input and front end. One example is a complex baseband window of 1024 samples (1024 μs at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control. The spectral front end applies a two-sided STFT with $N_{\text{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospective capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4} , weight decay 10^{-2} , batch

size 16, mixed precision, gradient-norm clipping at 5.0, and a shared 30-epoch budget. Auxiliary losses follow a fixed staged schedule, after which the learning rate decays on a cosine to a floor of 0.05 times its initial value. The reported checkpoint is chosen solely by validation modulation cross-entropy under a common eligibility window and 8-epoch early-stopping rule. No comparator received an architecture-specific sweep, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 bundle but not isolated.

Seeds. The sealed campaign uses 10 algorithm seeds shared across all models, with seed-aligned initialization; matched-subset displays use the first five. Schedules, seeds, teacher construction, and cache provenance are recorded in the reproduction package.

IV. EXPERIMENTAL DESIGN AND STATISTICAL ANALYSIS

A. Source-Disjoint Simulation Campaign

Macro-F1 is the primary score; it gives each modulation class equal weight, including classes with fewer examples. The fixed campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. SNR measures target strength relative to receiver noise, whereas SIR measures target strength relative to the jammer; more negative SIR therefore

TABLE I
COMPARATOR AND COMPLEXITY SUMMARY (HARD-INTERFERENCE
SETTING).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

means stronger interference. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans $\text{SIR} \in \{-15, -10, -5, 0\}$ dB, so $\text{SIR} = -15$ dB occurs only in the hard-interference split and $\text{SIR} = +5$ and $+10$ dB only in the in-distribution split. Each hard-interference SNR–SIR evaluation cell of Fig. 2 contains 131–181 source-disjoint test windows (mean 156; all ten classes, 5–26 per class), read from the frozen prediction cache. The envelope aggregates of Section V-D use a coarser partition of the same predictions (216–279 source-disjoint windows per held cell, mean 250). The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package but does not enter any sealed, envelope, or reported selector claim.

B. Comparator Fairness

Table I summarizes the comparators. All models are retrained under the same source-disjoint protocol; “IQFormer-inspired” and “MCLDNN” denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as a supervised adaptation reference; its 8.63 million parameters but only 155.3 million MACs reflect aggressive encoder downsampling and a large projection head, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator “IQFormer-insp.”; “A5” and “VIMD-Net” denote the same sealed configuration, and “CSSL” abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [36] we retrain under our protocol. Because this campaign retrains CSSL under the common supervised budget without executing the released self-supervised pretraining stage, it is reported as a supervised adaptation reference rather than a reproduction of the full released recipe.

As a public-benchmark comparator-fidelity check, the MCLDNN and IQFormer implementations were additionally evaluated on the public RML2016.10a benchmark [1] under the official IQFormer-release benchmark protocol [37]: the

11-class set, the full -20 to 18 dB SNR grid, per-stratum 600/200/200 splits seeded at 233, no augmentation or input-length change. Over three algorithm seeds, all-SNR test accuracy is 58.55% for MCLDNN (seed range 58.1–59.4%) and 65.23% for IQFormer-inspired (64.8–65.8%; seed ranges, not confidence intervals); the reference-release values, averaged over the same twenty SNR bins, are 62.05% and 64.19%. IQFormer-inspired closely matches the reference-release all-SNR accuracy ($+1.04$ pp), whereas MCLDNN remains moderately lower (-3.50 pp) while staying in the same broad all-SNR range. Per-SNR deviations of up to about 8 pp remain, so E1 speaks only to range-level comparator-implementation fidelity.

C. Baseline Configuration Contrast

The original confirmatory family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A5 differs from A0 in allocation, teacher, auxiliary objectives, residual bypass, and capacity (about $4.7\times$ parameters), so A5–A0 is a whole-configuration effect.

D. Evidence Scope and Independent Validation

The three configuration contrasts were fixed before the campaign analysis. The SNR–SIR map, overlap envelope, fusion, and capacity studies are exploratory analyses of the campaign predictions, including the post-hoc -15 -dB SIR stratification. They describe where the ranking changes but do not establish that the same boundary will hold in a new regime.

That transport question is evaluated on an independently generated severe-held set. The transfer analysis was specified before inference on that set. The sidecar analysis was specified after the transfer result but before sidecar evaluation, and the attribution controls were fixed before their respective executions. This sequence separates the original configuration contrast, campaign diagnosis, independent transport test, and subsequent repair intervention. The Supplement records the detailed execution timeline and reproducibility provenance.

E. Statistical Procedures

The original confirmatory family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, an exploratory calibration. Because seeds are fixed blocks, the reported intervals are conditional on the realized seed set; a seed-randomization variant resampling the ten realized seeds with replacement keeps the A5–A0 contrast positive (95% percentile interval $[4.02, 5.19]$ percentage points) while the teacher-form and route-count contrasts span zero ($[-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points). In the

TABLE II
ORIGINAL CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES
ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE
POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	-0.76	1.08
Tri-route vs dual-route	0.44	-0.47	1.36

campaign’s severe row, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples. The $\text{SIR} = -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2. The severe-held levels and repair contrasts of Section V-G instead average per-seed macro-F1 values with equally weighted regimes (the seed-wise paired convention); the two conventions are never mixed within a single reported contrast.

V. RESULTS

A. Compact Baseline Configuration

Table II reports the original confirmatory family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of [3.65, 5.49] percentage points, about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their simultaneous upper bounds are 1.08 and 1.36 percentage points, respectively. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution. Because the pre-specified joint criterion required all three contrasts to exclude zero, the joint criterion was not satisfied. The A5–A0 contrast nevertheless remains positive under the simultaneous family-wise interval. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Supplement Table IV); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design’s power; we attribute the compact frontier position to the VIMD family at this budget as a whole.

B. Condition-Dependent Ranking within the Campaign

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired; the averages favor the larger models. Figure 2 shows why no single campaign-level ranking is adequate: the sign changes with SNR and SIR, including a severe-row reversal. Section V-C tests whether this campaign structure transfers to independently generated severe regimes.

At $\text{SIR} = -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ([7.52, 12.14]) and 10.46 percentage points over IQFormer-inspired ([6.78, 14.29]); these

two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. Together, these checks show that the severe-row reversal is internally consistent across seeds and cells, while its interpretation remains campaign-specific. Section V-C evaluates this interpretation on independently generated severe-held data, where the favorable ordering does not persist. The reversal against MCLDNN is not explained by a binding training cap: MCLDNN early-stops in all ten seeds with best epochs well inside the shared budget (median 13, range 11–16; Supplement Table VIII).

C. Independent Severe-Regime Transport Boundary

An independently generated severe-held set tests whether the campaign’s -15 -dB ordering survives broader regime changes. SIR remains fixed at -15 dB: R1 combines held jammer families with 180-km/h mobility over training channels, whereas R2 combines held channels with the same jammer families at 250 km/h. Relative to the campaign corner, R1 changes jammer family and speed; R2 changes jammer family, speed, and channel. Each regime spans eight SNRs and 6400 source-disjoint examples (12800 total), and all evaluated checkpoints remain fixed.

The campaign ordering does not persist. Under the seed-wise convention, IQFormer-inspired and MCLDNN lead A5 by 2.58 and 13.03 percentage points, respectively; the corresponding window-pooled leads are 3.12 and 12.45 percentage points. The campaign-derived overlap–SIR diagnostic surface of Section V-D also loses transport skill: its sign accuracy on the severe-held cells falls to 0.578 against IQFormer-inspired and 0.406 against MCLDNN, while SIR-only or constant predictors outperform the full surface. Thus, the campaign’s favorable compact-model corner and its diagnostic boundary do not define a transferable severe-regime ordering.

The held jammer families clarify the remaining performance levels. Under OFDM-like interference every evaluated model is near the floor (4.60% for A5), whereas the pulse family remains learnable (29.03% for A5 and 55.34% for MCLDNN). This difference also means that jammer-family difficulty and out-of-domainness are confounded in the joint transfer result. Per-family and per-regime levels are provided in the Supplement.

D. Campaign Diagnostic and Its Transport Limit

The campaign-internal rank instability motivates a low-dimensional diagnostic surface that summarizes how gain varies with target-support jammer overlap and SIR. For reference b , we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b}o + \beta_{2,b}\text{SIR}, \quad (8)$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*. We avoid the more common “spectral occupancy” because o is not an occupied-bin or occupied-bandwidth fraction. The covariate is defined from the simulator-tracked target and jammer components rather

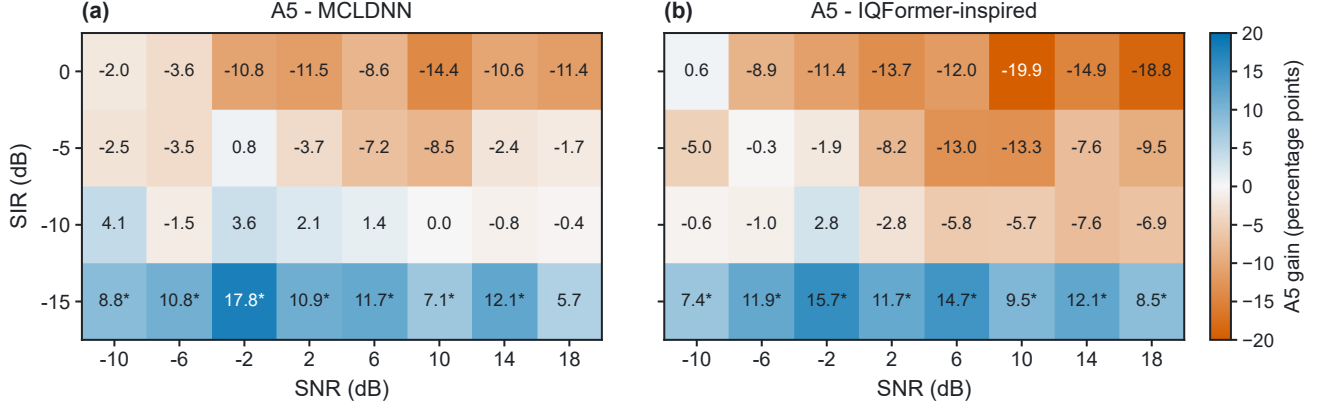


Fig. 2. Campaign-internal post-hoc rank instability over all hard-interference SNR-SIR cells. Each entry is A5 minus the named I/Q reference in macro-F1 percentage points. Blue cells favor A5 and orange cells favor the reference; color and printed values encode the same sign. A star marks a positive unadjusted paired lower interval; Holm-surviving cells are reported in the text.

than from the mixture. For each window we compute the STFT power of the tracked target and of the tracked jammer separately, order the time-frequency cells by descending target power, and take the target’s 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer’s total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t) \in \mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \quad (9)$$

Thus $o = 0.8$ means that 80% of the jammer’s energy lies within the target’s dominant support, not that the jammer fills 80% of the plane. The cell-level covariate is the mean of o over the windows of that cell.

The overlap is computed from simulator-tracked components on a fixed analysis lattice before model training; it is neither a receiver input nor a quantity available to an uncooperative receiver. The fit uses 32 cells from the in-distribution and hard-interference splits, and evaluation uses 80 cells from unseen-jammer, unseen-speed, held-channel, and combined-OOD splits. Each cell retains all ten classes and at least 150 source-disjoint windows. Complete cell construction and sensitivity checks are provided in the Supplement. Equation (8) is interpreted as a descriptive summary rather than a causal model.

Against IQFormer-inspired, the surface reduces held-cell RMSE from 8.37 to 6.23 percentage points ($R_{\text{skill}}^2 = 0.45$); against MCLDNN the corresponding values are 9.63 and 8.10 percentage points ($R_{\text{skill}}^2 = 0.29$). This pooled skill is regime-dependent, and paired tests do not resolve an incremental overlap effect beyond SIR. More importantly, leave-15-dB-out tests and the independent severe-held evaluation fail to support the full surface over simpler SIR-only or constant predictors. Figure 3 therefore visualizes a useful campaign summary, not a transferable switching rule.

The fitted break-even references and the unexpected positive overlap direction remain simulator-domain observations. Because the directional expectation was rejected and overlap is not directly observable at the receiver, these quantities are retained only as diagnostic evidence.

E. Complementarity and Cost Motivation

On the campaign hard-interference split, late probability fusion improves over the stronger constituent by 4.71 percentage points with IQFormer-inspired and 5.70 percentage points with MCLDNN. Same-model two-seed ensembles improve by 1.49 and 1.69 percentage points. Computed from the unrounded values, the corresponding excess cross-model fusion gains are 3.22 and 4.02 percentage points. This difference is consistent with complementarity beyond ordinary same-model ensembling. The diagnostic requires concurrent execution of both models and was not evaluated as the severe-held repair mechanism.

Table I gives the corresponding parameter and MAC budgets; compactness is stated as a parameter and MACs property.

F. Diagnosing the Representation Limitation

The cross-regime severe degradation makes representation robustness central to interpretation. The earlier clean-condition diagnosis independently tests whether the compact spectral representation discards information that becomes important under distribution shift. The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 4, where a star denotes a class whose jammer-free condition was not seen during training. Transfer failure occurs in 27/27 uncovered compact-spectral class-model cells (100%) versus 1/9 I/Q-reference cells (11%). The clean-retention gate did not pass. Figure 4 localizes the failure at the class level.

We then followed a prospective decision chain. Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation; probe performance is the unweighted mean F1 over QPSK, 16QAM, and 64QAM (chance ≈ 0.1 on ten classes). The predefined decision rule declares a representation *information-present*, *head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise, requiring

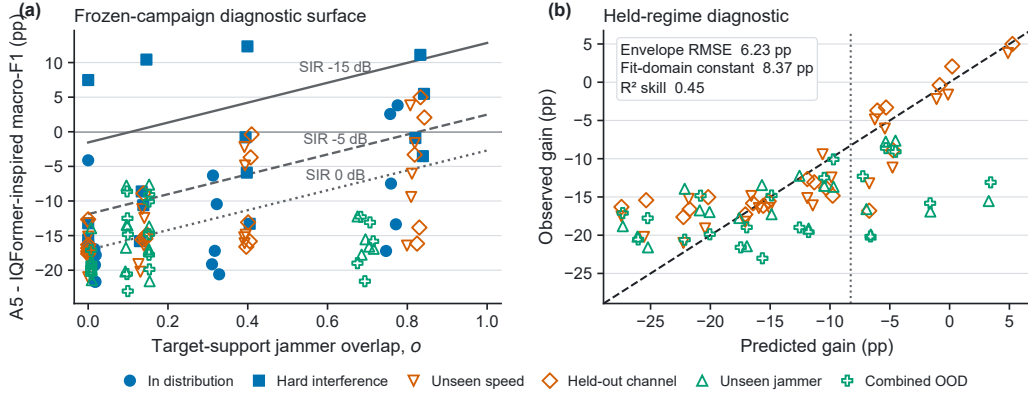


Fig. 3. Campaign-diagnostic overlap-SIR envelope against IQFormer-inspired. Panel (a) uses the target-support jammer overlap, not occupied bandwidth: filled markers are the 32 fit cells and open markers the 80 held cells. Panel (b) compares predicted and observed held-cell gain: RMSE 6.23 percentage points versus 8.37 percentage points for the frozen fit-domain constant ($R^2_{\text{skill}} = 0.45$). The surface summarizes campaign-internal rank variation but does not transport to the independent severe-held regimes of Section V-C.

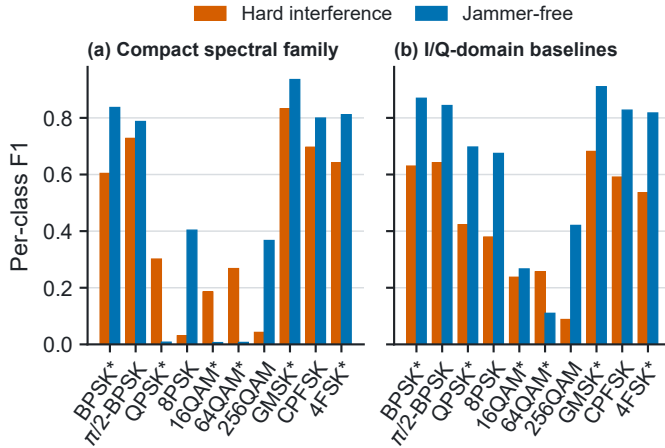


Fig. 4. Clean condition-transfer taxonomy. Bars show family-mean per-class F1 under hard interference (hatched orange) and jammer-free evaluation (blue) for the compact spectral family and I/Q-domain baselines. A star marks a class whose jammer-free condition was absent from training; it is not a significance marker.

all design-seed cells to exceed the threshold. For A5, five of the six design-seed cells (three algorithm seeds $\times$ the reconstructed jammer-free counterfactual and clean cross-validation probes) lie below the 0.25 threshold; the sole cell above it is borderline at 0.256. Because the decision changes under a 0.20 threshold, we treat the probe result as directional evidence for representation limitation rather than as a threshold-robust classification of mechanism. The strong I/Q baselines exceed the threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A predefined negative control gates the modulation route using predicted interference presence. Relative to

A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, saturating near the modulated route; this arm serves as a negative control.

G. Low-Cost I/Q Repair and Attribution

Widening the spectral model without changing its representation domain first separates capacity from representation. The widest spectral tier remains below IQFormer-inspired on the campaign hard split (L-IQFormer-inspired: -2.84 percentage points $[-4.30, -1.31]$) and on clean retention (-18.14 $[-19.34, -17.00]$ percentage points). At SIR = -15 dB it gains only $+4.96$ percentage points ($[3.48, 6.48]$) over the sealed tier. Tested spectral width scaling alone therefore does not remove the representation limitation.

The sidecar pools mean, standard deviation, and maximum statistics from a two-channel depthwise-separable received-I/Q branch, fuses its 24-dimensional embedding residually with the spectral path, and adds a side head. It uses only the received mixture and adds about 3% compute over A5. On the original campaign, the same configuration gains 11.77 percentage points on clean retention and 4.37 percentage points on the hard split relative to A5.

On the same frozen severe-held benchmark that exposed the transport boundary, the prospectively specified sidecar raises seed-mean pooled macro-F1 from 20.72% to 25.09%, a paired gain of 4.37 percentage points ($[2.96, 5.90]$). Repair is positive in all 10 seeds and all 8 SNR strata. Absolute macro-F1 remains low for every model at this extreme operating point, so the gain is interpreted as robustness repair. Coincidentally, the campaign hard-split sidecar gain is also 4.37 percentage points, but it is a distinct ten-seed measurement with interval $[3.44, 5.31]$ percentage points. Because this domain had already been characterized by the preceding transfer test, the sidecar experiment evaluates a prospectively specified intervention on a known held domain rather than an independent replication of repair transport.

TABLE III

TRANSPORT, REPAIR, AND ATTRIBUTION CONTROLS. POSITIVE CONTRASTS FAVOR THE ROW MODEL; WHERE REPORTED, CONFIDENCE INTERVALS ARE SEED-WISE PAIRED 95% INTERVALS. LESION ROWS MEASURE INFERENCE-TIME SENSITIVITY OF THE JOINTLY TRAINED SIDECAR. PANEL (C) REPORTS A FIVE-SEED VARIANT TRAINED ON THE ORIGINAL CAMPAIGN PARTITIONS WITH ITS I/Q INPUT PERMANENTLY ZEROED; SEVERE-HELD DATA REMAIN EVALUATION-ONLY.

(a) Full ten-seed severe-held evaluation				
Model or intervention	Macro-F1 (%)	Reference	Contrast	
A5 spectral baseline	20.72	—	reference level	
IQFormer-inspired	23.29	A5	+2.58 percentage points	
MCLDNN	33.75	A5	+13.03 percentage points	
Received-I/Q sidecar	25.09	A5	+4.37 percentage points ([2.96, 5.90])	
Sidecar, I/Q zeroed	12.80	full sidecar	−12.28 percentage points	
Sidecar, I/Q shuffled	16.25	full sidecar	−8.84 percentage points	
Sidecar, side head removed	7.09	full sidecar	−17.99 percentage points (−13.62 percentage points vs. A5)	
(b) Matched five-seed capacity control				
Measure	A5	Width M	Width L	I/Q sidecar
Macro-F1 (%)	20.72	21.77	20.89	26.11
MACs (M)	41.8	104.1	249.1	43.1
Seeds	5	5	5	5
(c) Retrained attribution ablation				
Condition	Macro-F1 (%)	Reference	Contrast	Seeds
Sidecar retrained, I/Q permanently zeroed	20.18	A5	−0.54 percentage points ([−1.34, 0.26])	5
Full sidecar (matched seeds)	26.11	A5	+5.39 percentage points ([3.54, 7.66])	5

The controls in Table III separate tested spectral width scaling from received-I/Q information and the added fusion head. On the matched five-seed severe-held subset, Width L gains only +0.17 percentage points over A5 at 249.1 M MACs, whereas the sidecar gains +5.39 percentage points ([3.54, 7.66]) at 43.1 M MACs. Thus, the larger repair has substantially lower incremental compute. The L tier’s 249.1 M MACs are already within the range of the large I/Q comparators (355.6 M and 398.2 M), so the width ladder does not retain the compute advantage that motivates the compact receiver. The remaining controls are inference-time lesions of the jointly trained sidecar; each lesion removes 2.02–4.12 times the magnitude of the repair effect, and the lesioned variants fall below the unmodified A5 baseline. They establish inference-time dependence on received-I/Q input but cannot quantify the branch’s information contribution. The follow-up retrained the same architecture on the original campaign training and validation partitions with the I/Q input permanently zeroed (five matched seeds and the same recipe); the severe-held set remained evaluation-only. The zeroed variant gains −0.54 percentage points versus A5 ([−1.34, 0.26]), while the matched full sidecar shows a gain of 5.39 percentage points ([3.54, 7.66]). The zeroed variant also remains at the A5 level on the campaign hard split, arguing against failed optimization as the explanation. Thus, within the tested architecture and simulation domain, the repair depends on informative received-I/Q input rather than on the added capacity and fusion structure alone.

VI. DISCUSSION

A. Implications for AMC Evaluation

The experiments show that model ranking is condition- and regime-dependent. Marginal scores favor the I/Q references,

the campaign contains a severe-condition reversal, and the independent severe-held regimes reverse that ordering again. The appropriate object of comparison is therefore the condition-dependent ranking, not a single global leaderboard.

This distinction changes the role of a compact-model gain. A favorable cell or stratum identifies where a representation is competitive inside the tested campaign; it does not establish that the same ordering will survive a joint change in jammer family, mobility, and channel. Reporting both marginal and condition-resolved rankings is therefore necessary when a learned receiver is selected for a variable operating environment.

B. Why the Campaign Diagnostic Does Not Transport

Equation (8) has skill over 80 held cells inside the frozen campaign, but that skill is heterogeneous by regime. The incremental overlap effect beyond SIR is not separately resolved, and leave−15-dB-out behavior depends on the reference. Gain also increases rather than decreases with overlap inside the campaign, opposite to the predefined directional expectation. The surface is therefore most useful as a descriptive diagnostic. Its overlap variable additionally requires simulator-tracked target and jammer components and is not directly observable at an uncooperative receiver.

The independently generated severe-held evaluation tests the campaign-internal severe ranking under new jammer, mobility, and channel combinations. The favorable ordering does not persist, defining a clear cross-regime transport boundary. This result also shows why the in-campaign break-even references should be treated as descriptive diagnostics rather than cross-regime switching thresholds.

C. Representation Repair and Receiver Design

The severe-held controls expose a compute-representation asymmetry within the tested architecture and regimes: substantial spectral widening does not reproduce the repair, whereas a small received-I/Q path produces a much larger gain under the reported MAC budget. Inference-time lesions show dependence on the I/Q path, and the retrained zeroed-I/Q control shows that added capacity and fusion structure without informative I/Q input do not reproduce the gain. Within the tested architecture, these results identify received-I/Q information as the factor enabling the repair. This identifies a mechanism-level effect within the tested simulation setting; deployment performance remains to be established.

At $f_c = 5.9$ GHz and the held-speed conditions, the channel can evolve materially within the 1024- μ s decision window: the nominal coherence time at 180 and 250 km/h is about 42% and 30% of the window, increasing intra-window nonstationarity. Representation robustness is therefore a receiver-design issue as well as a classifier-ranking issue. More generally, the results caution against fixing a learned receiver architecture or switching rule solely from one campaign's marginal leaderboard or condition surface. For compute-constrained learned receivers, the practical lesson is to test ranking stability across operating regimes and preserve a low-cost complementary signal path when a compact representation is vulnerable to shift.

VII. LIMITATIONS

The study has several limitations. First, all results are based on source-disjoint TR 38.901 TDL-profile simulations and do not constitute over-the-air or MAC-/network-layer V2X validation. Second, the confirmatory family resolves only the A5-A0 whole-configuration contrast; the isolated teacher-form and route-count effects remain unresolved. Third, the severe-held regimes change jammer family together with mobility, and R2 also changes channel profile. The transfer result therefore bounds joint transport and does not identify a single causal shift factor. Fourth, IQFormer-inspired and CSSL reach the shared 30-epoch limit in all seeds, so their comparisons retain a budget-sufficiency caveat; MCLDNN early-stops in all seeds and is not subject to the same immediate cap. Fifth, the overlap covariate requires simulator-tracked target and jammer components that are unavailable at an uncooperative receiver. In addition, absolute macro-F1 remains low in the extreme severe-held regime. The RML2016.10a check addresses comparator-implementation fidelity rather than validation of the proposed representation. RML2016.10a does not expose the separately tracked post-channel target, jammer, and residual components required by the training-only teacher, so it cannot instantiate the full A5 training recipe.

Reproducibility

The artifact package preserves the source data, cache identities, checkpoints, analysis definitions, figure-source CSVs, per-seed aggregates, configurations, and provenance associated with each evidence stage. Each manuscript macro stores its source path, row selector, evidence class, precision, and source SHA-256. Together, these records support direct reproduction

of the reported analyses; the resolved package will be released upon acceptance.

VIII. CONCLUSION

A single marginal AMC score is insufficient when the preferred signal representation changes across interference and mobility regimes. The reported campaign reveals such a rank reversal, while independent severe-held evaluation shows that neither the favorable compact-model corner nor its overlap-SIR diagnostic transfers uniformly to new regime combinations. A lightweight received-I/Q sidecar recovers 4.37 percentage points on that benchmark at about 3% additional MACs. Spectral widening does not reproduce the gain, and a retrained zeroed-I/Q variant returns to the spectral baseline, supporting informative I/Q input as the enabling factor within the tested architecture. Because absolute performance remains low and all evidence is simulation-based, the result is a bounded robustness recovery. The broader receiver-design implication is to test ranking stability explicitly and retain a low-cost complementary signal path when compact representations are exposed to regime shift.

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